VALIDATION REPORT FOR

MYANMAR KINDAR SOLAR POWER PLANT PROJECT

CONTACT INFORMATION

Project Title	Myanmar Kindar Solar Power Plant Project
Version	01.1
Report ID	0104020323023(001)

Report Title	Validation Report for Myanmar Kindar Solar Power Plant Project
Client	Myanmar Kindar Solar Power Company Limited
Date of Issue	26/07/2024

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Summary:

China Classification Society Certification Co. Ltd. (hereafter referred to as "CCSC") has been commissioned by Myanmar Kindar Solar Power Company Limited to perform the validation of the project activity "Myanmar Kindar Solar Power Plant Project" (hereafter referred to as "the project") on the basis of requirements of Verified Carbon Standard (VCS) Version 4.5, VCS Validation and Verification Manual, and the approved Clean Development Mechanism (CDM) methodology ACM0002 version 21.0, as well as criteria given to provide for consistent project operations, monitoring and reporting. The validation report is summarizing the findings of the validation.

The project is located in the south of Mandalay Division and 56km northeast to the Meiktila town, Mandalay Division, Myanmar. The purpose of the project is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. The project installed 74,592 pieces of polycrystalline silicon PV module, with unit capacity 540 Wp, providing a total capacity of 40.28 MWp. The annual emission reductions in the crediting period is expected 22,402 tCO_2e. Installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately. The monitoring system is in place and the project reduces GHG emission. The GHG emission reduction is calculated without material misstatements, and the emission reductions verified totalize 156,814 tCO_2e for the 1st renewable crediting period.

The validation scope is defined as an independent and objective review of the VCS Project Description (hereafter refer to as "PD"), the project's baseline study and monitoring plan, the emission reduction calculation spreadsheet and other relevant documents refer to VCS and CDM requirements.

The validation is an independent 3rd party assessment of the project's baseline, estimated GHG emission reductions or net anthropogenic GHG removals, the monitoring plan and the crediting period using the applicable VCS requirements. Validation is a requirement for all VCS projects and is seen as necessary to

provide assurance to stakeholders of the quality of the project and its intended generation of estimated Verified Carbon Units (VCUs). In carrying out its validation work, the VVB shall ensure that the project complies the applicable CDM and VCS requirements.

The project is eligible under non-AFOLU project. The validation criteria followed the guidance documents provided by VCS and the selected CDM methodology and tools includes:

- VCS Standard, Version 4.5
- VCS Program Guide, Version 4.4
- Methodology ACM0002 version 21.0: Grid-connected electricity generation from renewable sources and related CDM tools

The Validation process includes three phases:

1) Desk review of documents;

2) Remote visit and follow-up interviews;

3) Issuance of the final validation report and opinion.

CCSC appointed a qualified audit team in accordance with internal procedures to perform the validation. Two Corrective Action Requests (CAR) and one Clarification Request (CL) were raised in the validation process and successfully closed upon the project participant taken actions and submitted the revised PD and supporting evidence. No Forward Action Request (FAR) was raised during this validation.

As CCSC internal QA/QC control procedures, the audit team has strictly carried out ISO 14064-3:2019, ISO 14065:2020 and the VCS requirements. Therefore, there are no restrictions of uncertainty for the validation.

In conclusion, it is CCSC's opinion that the project in the VCS PD (version 02, date 26/02/2024) meets all relevant requirements for VCS activities and all relevant host Party criteria and correctly applies methodology ACM0002 version 21.0: Grid-connected electricity generation from renewable sources.

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1 INTRODUCTION

1.1 Objective

Myanmar Kindar Solar Power Company Limited has commissioned China Classification Society Certification Co., Ltd. (herein after refer to as CCSC) to performed the validation of Myanmar Kindar Solar Power Plant Project (VCS ID 4580, hereafter referred to as "the project"), on the basis of requirements of Verified Carbon Standard (VCS) version 4.5.

CCSC as the validation/verification body (VVB) of the project has been accredited as a DOE by UNFCCC and meets the competence requirements as set out in ISO 14065:2020.

The validation is an independent 3rd party assessment of the project's baseline, estimated GHG emission reductions or net anthropogenic GHG removals, the monitoring plan and the crediting period using the applicable CDM and VCS requirements.

1.2 Scope and Criteria

The validation was performed on the basis of VCS criteria. The information in the documents is reviewed against CDM Validation and Verification Standard, VCS standard version 4.5 and associated interpretations. CCSC has based on the instructions in the VVS and employed a risk-based and stepwise approach when conducting the validation, focusing on the identification of significant risks for project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the PPs. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

1.3 Reasonableness of Assumptions

CCSC has undertaken a reasonable assurance engagement in accordance with VCS Standard (version 4.5). The validation and verification conclusions are based on the PD and supporting evidence made available to the VVB and information collected through performing interviews and during the remote inspection.

1.4 Summary Description of the Project

The project is located in the south of Mandalay Division and 56km northeast to the Meiktila town, Mandalay Division, Myanmar. Prior to the implementation of the project, electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. Baseline scenario is the same as the scenario prior the implementation of the project.

The purpose of the project is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. The project installed 74,592 pieces of polycrystalline silicon PV module, with unit capacity

540 Wp, providing a total capacity of 40.28 MWp. The annual emission reductions in the crediting period is expected 22,402 tCO_2e. The project was put into production since 18/01/2023.

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation scope encompasses an independent and objective review and ex-post determination of the monitored reductions in GHG emissions by the CCSC. The validation scope covers the relevant documents (e.g., the registered VCS PD, the emission reduction calculation spreadsheet, supporting documents and information collected through performing interviews and during the remote inspection, VERRA's requirements publicly available, relevant rules, including the host country legislation, etc.) to be independently reviewed, the project geographical locations to be visited remote, the related project stakeholders to be interviewed with, and processes that are necessary to acquire objective evidence for the evaluation of the project compliance to the VCS requirements and associated interpretations. The validation activities are conducted according to the VERRA's requirements. In doing so, the principles of accuracy and completeness, relevance, reliability and credibility were followed. The validation is not meant to provide any consulting service towards the PPs. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project.

The validation schedule including key milestones has been included in below of this table.

Table 2.1.1 the validation and verification schedule

Item	Date	Description of the implementation
Desk review of documents	11/12/2023 to 15/12/2023	The audit team evaluates the information provided by PPs.
Opening meeting	23/02/2024	All parties involved had attended the kick-off meeting
Remote inspection	23/02/2024	The audit team has interviewed with key personnel from the project owner (the project proponent) and buyer & developer.
The resolution of outstanding issues and the issuance of the validation report and statement.	23/02/2024 to 26/02/2024	After the remote inspection, The audit team issuance of the findings and subsequently the final validation report.

The validation activities are conducted according to the VERRA's requirements. In doing so, the principles of accuracy and completeness, relevance, reliability and credibility were followed. The validation is not meant to provide any consulting service towards the PPs. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project.

2.2 Document Review

A desk review of the VCS PD and supporting documents was conducted by the audit team. The aim of the desk review of the documentation was to verify correctness, credibility, interpretation and completeness of presented data and information, and to cross check between information provided in the updated VCS PD and information from sources other than that used, if available. Review of the appropriateness of formulae and correctness of calculations was also carried out during this stage based on the approved methodology being applied. And particular attention was given to the quality of the metering equipment including calibration requirements, the monitored data and its evidence, and the emission reduction calculation. The evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions and VCUs was also conducted.

The audit team evaluated the information provided by PPs through document review in accordance with the requirements of CDMP0310- CDM Audit Verification Implementation Procedure and CDM Project Audit Verification Standards, both of which are the internal procedures for implementing CDM/VCS Validation/Verification in CCSC. With the document review, the audit team should acquire including:

- Identification of project types;
- Description of project activities;
- Selection of methodology and baseline;
- Application of Methodologies and Tools;
- Start date, type of credit period and length of credit period;
- Environmental impact;
- Local stakeholders communication methods;
- Evaluation comments from local stakeholders;
- The state of sustainable development;
- Approval and authorization.

When assessing the project baseline, the audit team consult both the provided documents by PP and relevant national law and industry regulatory texts or refer to industry experts to ascertain the waste and waste treatment requirements of local regulations. This procedure enables the team to determine the accuracy of the project baseline selection.

Appendix 3 of this report contains a complete list of all documents and proofs reviewed by the audit team.

2.3 Interviews

On 23/02/2024, CCSC performed remote inspection against the physical site of the project.

During the remote inspection, the audit team has interviewed with key personnel from the project owner (the project proponent) and buyer & developer, and local stakeholders.

The assessment content and topics/the persons interviewed at the remote inspection were provided in the below table.

Date: 23/02/2024	
Interview topics	Interviewed Organizations and persons
Project background information and VCS consideration, such as: - Situations prior to the project implementation. - Participation under Other GHG Programs, etc. - Project approval status. - Energy supply and demand status. - Local policies. Stakeholder consultation process Social and environmental impact, SDG contribution, Stakeholders' comments - Social and environmental impact of the project, environmental monitoring. - Environmental protection requirements related to Project construction (engineering construction) and operation. - Stakeholders' comments. - SDG contribution of the project activity. Whether Myanmar has a market-based tool like ETS in the European Union aiming at GHG emission reductions. Will Myanmar establish one in the near future?	Myanmar Kindar Solar Power Company Limited (Project Owner): Mr. Li Yanjun Mr. Jing Yu Mr. Tang Chengkai Mr. Yang Ronghua Mr. Wuma Local residents Mr. An Kgaw Mr. Myo Myint Ko Mr. Kywa Ye Maung Ms. Ma May Lay Ms. Ma Hnin Mr. Soe Thee Oo Ms. Khaing Then Swe Mr. Min Min Zaw
- Applicability of selected methodology. - Baseline of the project. - Emission factors. - Preparation of Joint PD&MR. - Compliance of the monitoring plan with the monitoring methodology. - Compliance of monitoring with the monitoring plan. - Assessment of data and calculation of GHG emission reductions.	Myanmar Kindar Solar Power Company Limited (Project Owner) Mr. Li Yanjun Mr. Jing Yu Mr. Tang Chengkai Mr. Yang Ronghua Mr. Wuma

Date: 23/02/2024	
Interview topics	Interviewed Organizations and persons
- Monitoring devices. - Monitoring method.	
- The process and participation of the stakeholder consultation - The environmental and social imparts of the project - Any complaints by the local stakeholders and the implementation of the mitigation measures	Local stakeholders Mr. An Kgaw Mr. Myo Myint Ko Mr. Kywa Ye Maung Ms. Ma May Lay Ms. Ma Hnin Mr. Soe Thee Oo Ms. Khaing Then Swe Mr. Min Min Zaw

2.4 Site Visits

Subject to the security situations around the project site, the PP applied for the exemption of on-site inspection, which has been approved by VERRA on 20/02/2024, and the letter of approval of the exemption of on-site inspection can be found on the VERRA Registry page: https://registry.verra.org/app/projectDetail/VCS/4580. Therefore, CCSC validation team performed the remote inspection. Besides, the PP provided the pictures of facilities and the monitoring equipments etc, and finalised the questionnaire designed by the validation team. The PP also provided the Statement to declare authenticity of all the documents/videos/photos.

CDM-EB 113, Annex 7 "Amendments to version 03.0 of the CDM validation and verification standard for project activities on remote validation or verification by DOEs" specifies the "Guidance on remote inspection as an alternative means to an on-site inspection", which was followed by the validation team to conduct the remote inspection and all the risk included this guidance were considered:

(a) Risk assessment stage: i) Internet has been run smoothly both in PP and validation team's office for a long period and quality of the internet connection is fine. The information and communication technology (ICT) tool named Tencent Meeting was chosen for the communication between the validation team and the PP, since Tencent Meeting is allowed to connect internet both in Myanmar and China. The quality of good camerawork of the Tencent Meeting ensures a reasonably good view for the validation or validation team. The validation team estimated the amount of documentation to be reviewed remotely and concluded that the document review can be finished in time, and relevant data flows can be accessed remotely. Confidentiality rules were declared to the PP by the validation team which is in accordance with the CCSC's regulations, and competence and resources of the validation team ensure the validation to finish the remote inspection. ii) This project is a solar PV power project, and the boundary and features of the project activity which are specified in the methodology and the registered PDD can be evaluated remotely. The project does not involve any project emissions as per the methodology and the registered PDD. Crosscheck method for the monitored data is based on the document review. iii) The validation team studied the complexity of the monitoring parameters and the monitoring plan, data processing and reporting, and data and information flow, and concluded that the aspects of monitoring activities can be assessed thoroughly through remote inspection. iv) The validation team established measures to eliminate or reduce the identified risks. For example, if Tencent Meeting cannot ensure smooth communication, Microsoft Teams can serve as the alternative ICT tool; if the quality of the wireless network is not good, personal hotspot from the internet of the mobile phone will be an alternative.

(b) Planning stage: The validation team planned the validation and provided this plan to the interviewed PP. This validation plan includes the composition of the validation team with sufficient competence, the plan of desk review to gain a prior understanding of records and documentation control processes, tasks to be done during the remote inspection, determination of ICT tools to be used (Tencent Meeting).

(c) Implementation stage: During the remote inspection, the validation team implemented measures which has been established to mitigate the identified risks, and actual risks were not higher than initially assessed. The remote inspection has been successfully finished.

(d) Post-remote inspection stage: The validation team decided that another round of remote inspection is unnecessary while reviewing the project participant's response to clarification requests, corrective action requests and/or forward action requests. The technical review process is able to identify any risks that were not identified during risk assessment stage.

By the remote inspection and remote interview plus the document review, the CCSC validation team confirms the alternative measure used and the information provided by the PPs are credible and sufficient for the purpose of validation.

The audit team performed the remote inspection during 23/02/2024. The interviewed personnel and objective are listed in above table. The remote inspection involved following activities:

- An assessment of the implementation and operation of the project as per the PD;
- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the approved monitoring plan.

2.5 Resolution of Findings

During the validation of a project activity, to identify issues that need to be further elaborated upon, researched, or added to in order to confirm that the project meets the VERRA's requirements and can achieve credible emission reductions, CCSC will issue a Corrective Action Request (CAR), a Clarification Request (CL) or a Forward Action Request (FAR) depending on different situations.

The objective of this phase of is to resolve issues related to the monitoring, implementation and operations of the project activity that could impair the capacity of the project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions prior to CCSC's positive conclusion on the GHG emission reduction calculation.

Findings established during the validation and verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

A Corrective Action Request (CAR) will be raised if one of the following occurs:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project proponents, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project proponents;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project proponents.

A Clarification Request (CL) will be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A Forward Action Request (FAR) will be raised, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

There are 2 CARs, and 1 CL that have been raised during the validation process. No Forwarded Action Requests (FAR) were raised.

Each finding, including the issue raised, the response(s) provided by the project proponent, and the final conclusion and any resulting changes to project documents, are listed in the appendix 4.

2.5.1 Forward Action Requests

None FAR was raised during the validation & verification process.

3 VALIDATION FINDINGS

3.1 Project Details

The project is located in the south of Mandalay Division and 56km northeast to the Meiktila town, Mandalay Division, Myanmar. Prior to the implementation of the project, electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. Baseline scenario is the same as the scenario prior the implementation of the project.

The purpose of the project is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. The project consists of the installation of 74,592 pieces of 540 Wp polycrystalline silicon PV module, providing a total capacity of 40.28 MWp. The expected net annual power supply to the Myanmar Power Grid is 62,015.75MWh. The expected annual full load utilization hours is 1,540h corresponding to a PLF of 17.58%.The technical specifications of the main equipment of the project is shown in Table 3.1-1, which has been confirmed by reviewing the Technical specifications of main equipments /19/:

Table 3.1-1 Main Equipments Specifications

540 Wp polycrystalline silicon PV module	
Model	JAM72D30-540/MB
Peak power	540 Wp
photon-to-electron conversion efficiency	≥22.8
Size	2278*1134*35 (mm)
Life time	20 years
Manufacturer	JA Solar Technology Co., Ltd.
Inverter	
Model	SUN2000-215KTL-H0
Max input voltage	1500V
Rated output power	204kVA@40˚C
Max efficiency	⩾99%
Life time	20 years
Manufacturer	Huawei Digital Technology (Suzhou) Co., Ltd.

The project is expected to generate an annual average of 22,402 tCO_2e emission reduction over the crediting period.The project started construction in 05/2022 and was put into production since 18/01/2023.

Item	Evidence gathering activities, evidence checked, and assessment conclusion

Sectoral scope	Checking Annex A to the Kyoto Protocol: https://unfccc.int/resource/docs/convkp/kpeng.html and VERRA website: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes It is confirmed that the project belongs to sectoral scope 1: energy industries (renewable-/non-renewable sources) as the project is related to grid connected renewable (solar PV) power generation.
Project activity type, technologies and measures implemented	The purpose of the project is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. The project consists of the installation of 74,592 pieces of 540 Wp polycrystalline silicon PV module, providing a total capacity of 40.28 MWp. The expected net annual power supply to the Myanmar Power Grid is 62,015.75MWh.. By checking the Feasibility Study Report /13/ and remote inspection against the applied technology by the project, the audit team can confirm that the project activity type is grid connected renewable (solar PV) power generation.
Technologies and measures implemented by the project activity	The technology and measure implemented by the project activity is grid connected renewable (solar PV) power generation, which has been confirmed by remote inspection and document review /13//16/.
General eligibility of the project to participate in the VCS Program	• The project is grid connected renewable (solar PV) power generation project in LDC (Myanmar), which is not excluded under Table 2.1 of the VCS Standard. • The project start date is 18/01/2023, which is the operation start date confirmed by checking Electricity metering records /23/, while the validation has been complete within 2 years of the project start date, which complies with the VCS standard. • The applied CDM methodology ACM0002 version 21.0 is eligible under the VCS Program, and the methodology does not have scale and/or capacity limits. • No other relevant eligibility information.
Project ownership	The section 1.7 of the PD has been checked and it has been confirmed that the project proponent demonstrates that they have the legal right to control and operate project activity, via checking the Letter of Acceptance /12/, business license /11/, PPA /20/, and the

	contracts /15//16/ of equipments purchase and construction. Thus, it is confirmed that the project owner has the ownership of the project.
Project start date	The project start date 18/01/2023 is the date on which the project began generating GHG emission reductions or removals as per the concept in VCS Standard Version 4.5, which is confirmed by remote interviewing against the representatives of the PP and local stakeholders and checking the documents including Electricity metering records /21/.
Project crediting period	PP has chosen the renewable 7 years of crediting period, 18/01/2023 to 17/01/2030, which is in line with the concept for crediting period in section 3.9.1 of the VCS Standard Version 4.5.
Project scale	As the estimated annual average GHG emission reductions per year is 22,402 tCO_2e which is lower than 300,000 tCO_2e, the project classified as “Project” under VCS standard version 4.5. Through checking emission reductions calculation spreadsheets /3/ provided by PP, the audit team was able to confirm the results of the estimated GHG emission reductions of the project activity.
Project location	The project is located in the south of Mandalay Division and 56km northeast to the Meiktila town, Mandalay Division, Myanmar. The geographical coordinates are 96°19’19’’-96° 19’53’’E, 21°10’26″-21°11’03’’N. The location of the project site was verified through remote inspection and the coordinates were verified remote by GPS devices and by checking against Google Maps.
Conditions prior to project initiation	Prior to project initiation, electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. This has been confirmed by reviewing the documents including Feasibility Study Report /13/, and interviewing against the representatives from the local government and the PP.
Project compliance with applicable laws, statutes and other regulatory frameworks	By checking the government policies and interviewing the local experts, and review relevant documents including Business license of the project owner, Feasibility Study Report, Environmental Management Plan Report and etc. /11//13//14/, the audit team can confirm that the project meets the laws, statutes and other regulatory frameworks.

Double counting and participation under other GHG programs	The project has neither been registered nor seeking registration under any other GHG programs, and hasn't been rejected by any other GHG project, which has been confirmed via checking the website of UNFCCC, GS, VCS, GCC, and CCER.
No double claiming with emissions trading programs or binding emission limits	The audit team confirms that the Net GHG emission reductions or removals generated by the project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. The audit team checked the relevant laws, statutes, and other regulatory, and found that there is no market-based tool like ETS in the European Union aiming at GHG emission reductions in the host country. Thus, the audit team concluded that the project activity will not be involved on other Emissions trading programs and other binding limits.
No double claiming with other forms of environmental credit	The project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program, evidenced by the declaration /25/.
Supply chain (Scope 3) emissions double claiming	The audit team confirmed that since the project utilizes solar power for grid electricity generation, it does not impact emissions associated with a good or service. Therefore this section is not applicable.
Sustainable development contributions	By interviewing with stakeholders during remote interview, it is confirmed that the project activity would contribute sustainable development in the region in following aspects: - 7.1.1 Proportion of population with access to electricity - 8.5.1 Average hourly earnings of employees - 13.3 Tonnes of greenhouse gas emissions avoided or removed
Additional information relevant to the project	• Leakage management for AFOLU projects According to the methodology, the leakage of the project is zero, thus the leakage management is not applicable to the project. • Commercially sensitive information No commercially sensitive information has been excluded from the public version of the project description. • No additional relevant information that may have a bearing on the eligibility of the project, the reductions or removals, or the quantification of the project's

	reductions or removals.

Therefore, the audit team is able to conclude that the description in the project description is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

The project purpose is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. There are no harms identified from the project.

3.2.2 Local Stakeholder Consultation

The audit team has checked the PD and confirm that the stakeholder identification has been conducted as per the VCS standard Version 4.5. Local stakeholders mean the public, including individuals, groups or communities, affected, or likely to be affected, by the project. Regarding the project the local stakeholders include the local residents and communities.

By reviewing the Environmental Management Plan (EMP) /14/ and interviewing the representatives of the project owner and the local stakeholders, public consultation meeting concerning with the for the construction and operation of the project was held on 22/05/2022 during the Environmental Impact Assessment process in order to collect local stakeholders' comments. According to the Environmental Management Plan (EMP) /14/, the project owner invited local people by negotiating with village administrators. In total 23 attendees attended this public consultation meeting. Most of the local stakeholders knew about solar PV power projects and all of them held positive and supportive attitude towards the construction of the proposed project.

By interviewing the representatives of the project owner and the local stakeholders, the audit team confirmed that the project owner has set out the mechanism for on-going communication with local stakeholders and the communications with local stakeholders are carried out in any time. The project owner published the contact information of the contact person who is responsible for stakeholders comments. Stakeholders were informed of the contact information, and their comments can be directly collected by the contact person. The comments would be fed back to the stakeholders by the contact person for a timely response. Once the contact person received negative comments from the stakeholders, the contact person would record the negative comments and the feedback.

3.2.3 Environmental Impact

By reviewing the Environmental Management Plan (EMP) /14/, the audit team confirmed that the environmental impacts of the solar PV project during the construction period and operation period including

1. Impact on air environment

2. Impact on water environment

3. Noise and Vibration Impact

4. Impact on terrestrial ecology

5. Solid Wastes Generation Impact

All the impacts are considered insignificant on local environment, which is confirmed by the audit team through reviewing the Environmental Management Plan (EMP) /14/.

3.2.4 Public Comments

The assessment team noted that this project was open for public comment from 21/09/2023 to 21/10/2023. The detail was checked by the assessment team in the following web platform:

https://registry.verra.org/app/projectDetail/VCS/4580.

During the period, no public comments were received.

3.2.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project.

For non-AFOLU projects, this section is not required.

3.3 Application of Methodology

3.3.1 Title and Reference

Applied methodology: ACM0002 version 21.0 Grid-connected electricity generation from renewable sources

Applied tools:

Version 07.0.0 of Tool01 “Tool for the demonstration and assessment of additionality”.

Version 07.0 of Tool07 “Tool to calculate the emission factor for an electricity system”.

Version 03.1 of Tool24 “Common practice”.

Version 13.0 of Tool27 “Investment analysis”.

3.3.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002 version 21.0	1. This methodology is applicable to grid-connected renewable energy power generation project activities that:	Applicable as the project utilizing solar photovoltaic energy to supply

	(a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or Involve a replacement of (an) existing plant(s)/unit(s).	electricity to the Myanmar Power Grid, which is a Greenfield power plant.
ACM0002 version 21.0	2. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic[1] or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	N/A, as the project does not involve the integration of a BESS.
ACM0002 version 21.0	3. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the	Applicable. The project is a greenfield solar PV project without the integration of BESS.

[1] In case of retrofit or capacity addition for concentrated solar power projects, stakeholders may submit a request for revision to this methodology, providing an apportioning approach to calculate the project emissions due to any fossil fuel consumption attributed to the increased electricity generation from the BESS.

	start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies[2] may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 of the methodology. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
ACM0002 version 21.0	4. In case of hydro power plants, one of the following conditions shall apply:[3] (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m^2; or (c) The project activity results in new single or	N/A. The project is not a hydro power project.

[2] For example, upon deep discharge of the batteries.

[3] Project participants wishing to undertake a hydroelectric project activity that results in a new reservoir or an increase in the volume of an existing reservoir, in particular where reservoirs have no significant vegetative biomass in the catchments area, may request a revision to the approved consolidated methodology.

	multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
ACM0002 version 21.0	5. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	N/A. The project is not an integrated hydro power project.
ACM0002 version 21.0	6. The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of	N/A. The project does not involve switching from fossil fuels to

	the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	renewable energy sources at the site of the project activity, and is not a biomass fired project.
Version 07.0.0 of Tool01	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM for the project.
Version 07.0.0 of Tool01	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.
Version 07.0.0 of Tool01	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	N/A as the project is not a CDM project.
Version 07.0.0 of Tool01	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	N/A as the project does not involve biofuels.
Version 07.0 of Tool07	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose	Applicable as the methodology refers to this tool.

	alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	
Version 07.0 of Tool07	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Applicable as the methodology refers to this tool.
Version 03.1 of Tool24	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Since the project activity is using "Tool for the demonstration and assessment of additionality", the tool is applicable.
Version 03.1 of Tool24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology refers to "Tool for the demonstration and assessment of additionality". It required that the latest version of the "Methodological tool: Common practice" available on the UNFCCC website shall be applied. Therefore, this tool is applicable. Applicability condition is met.
Version 13.0 of Tool27	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Applicable since the project applies the methodological tool "Tool for the demonstration and assessment of additionality".
Version 13.0 of Tool27	In case the applied approved baseline and monitoring methodology contains requirements	N/A as the methodology does not contain requirements for

	for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	the investment analysis that are different from those described in this methodological tool.

3.3.3 Project Boundary

The audit team has checked the documents including Feasibility Study Report /13/, Technical specifications of main equipments /19/, and remote inspected the project and interviewed the PP, and confirmed that the project boundary is appropriately correctly in the PD which is in accordance with the methodology ACM0002 version 21.0. Emissions sources included or excluded from the project boundary for determination of baseline and project emissions are justified in accordance with the methodology.

3.3.4 Baseline Scenario

The project is the installation of a Greenfield power plant without a BESS. Baseline scenario has been identified in the PD which has been confirmed in accordance with the applied methodology ACM0002 version 21.0:

Electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “Tool to calculate the emission factor for an electricity system”.

3.3.5 Additionality

The PD of the project demonstrates the additionality using the Tool01 as per the methodology ACM0002 version 21.0. Step 1 ~ 4 of the Tool01 are followed, which are validated as follows:

Step 1. Identification of alternatives to the project activity consistent with current laws and regulations

<u>Sub-step 1a: Define alternatives to the project activity</u>

The PD identified realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity. The alternatives that are prima facie realistic and credible in the context of the Myanmar Power Grid include:

(a) The proposed project activity undertaken without being registered as a VCS project activity

(b) Continuation of the current situation: the same service of power supply provided from grid

It is confirmed that the above alternatives are in accordance with the description of the methodology.

<u>Sub-step 1b: Consistency with mandatory laws and regulations</u>

The audit team can confirm through document review /11//12//13//14/ that the project activity complies national policies for environmental protection, energy conservation and sustainable development, and there are no binding legal and regulatory requirements for this project type.

Outcome of Step 1: It is concluded that **scenario (a)** and **scenario (b)** are in compliance with the relevant laws and regulations. The following steps are used to demonstrate the additionality of the project according to the "Tool for the demonstration and assessment of additionality":

Step 2. Investment Analysis

As per the Tool0*1*, the objective of Step *2* is to *d*etermine whether the proposed project activity is not: (a) The most economically or financially attractive; or (b) Economically or financially feasible, without the revenue from the sale of emission reductions. *The PD takes into account the* latest approved version of the "Methodological tool: Investment analysis" (Tool27 version 13.0), available on the UNFCCC website following the Tool01.

Sub-step 2a: Determine appropriate analysis method

The Benchmark Analysis versus the Investment Comparison Analysis and Simple cost analysis is determined by whether the output can only be provided by the project proponent. In the PD, the analysis will be performed through Option III of the additionality tool, i.e. benchmark analysis. This method is applicable because:

- Option I: Simple cost analysis is not applicable since the project generates economic returns through the sales of electric power to the grid;
- Option II: Investment comparison analysis is only applicable when alternatives are also investment projects. However, the alternative baseline scenario (c) is the grid providing the same amount of power;
- Option III: Benchmark analysis is applicable as the return on investment relative to the industry benchmark was crucial for the decision to invest the project.

It is appropriate and valid that the PD choses the Benchmark Analysis method to demonstrate the additionality.

Sub-step 2b – Option III: Apply benchmark analysis

In the PD, benchmark analysis is conducted as per the CDM tool-27 "Investment Analysis". The equity IRR (Internal Rate of Return) is selected as the financial indicator of benchmark analysis to demonstrate additionality. The audit team checked the CDM tool-27 "Investment Analysis" and confirmed that according to the Appendix of the CDM tool-27, **20.98%** is the benchmark of the equity IRR in real term of Group 1 projects, which covers renewable power projects, in Myanmar. The audit team confirmed that the PP computed a conservative, which means a higher, nominal post tax project IRR for the project, since the fixed project costs are applied without considering the inflation.

As per the CDM tool-27 the post-tax equity IRR benchmark 20.98% in real terms should be converted to a benchmark in nominal terms. Since the inflation rate of Myanmar is above 0% in recent years[4], it is conservative to use the real term benchmark, 20.98% in the investment analysis, which is lower than the nominal benchmark, which is confirmed by the audit team.

Given the short period of time between the Feasibility Study Report (11/2021) /13/ and the construction start date of the Project in 05/2022, the audit team was therefore confident that it is unlikely in the context of the underlying Project that the input values would have materially changed. At the same time, the audit team compared the input values for the financial analysis in the PD and the Feasibility Study Report and confirms that all input parameters used in the financial analysis are taken from the Feasibility Study Report. Therefore, the audit team confirms that the input values from the Feasibility Study Report were valid and applicable to the investment analysis.

Furthermore, the audit team has reviewed the IRR calculation sheet and data sources of the parameters for investment analysis, and confirms that:

[4] https://www.statista.com/statistics/525770/inflation-rate-in-myanmar/

<table>
<tr><th>Parameter</th><th>Value</th><th>Unit</th><th>Source</th><th>Crosscheck</th><th>Justification</th></tr>
<tr><td>Total static investment</td><td>17,931.52</td><td>10,000CNY</td><td>Feasibility Study report</td><td>Actual contract values:
<table>
<tr><td>Contract</td><td>Value (CNY)</td></tr>
<tr><td>PV racking</td><td>954.12</td></tr>
<tr><td>PV module</td><td>7,935.10</td></tr>
<tr><td>Transformer</td><td>296.00</td></tr>
<tr><td>Engineering</td><td>9,053.17</td></tr>
<tr><td>Total</td><td>18,238.39</td></tr>
<tr><td>Percentage of estimated total investment</td><td>101.71%</td></tr>
</table></td><td>The available contracts sum up to 101.71% of the estimated total static investment. Therefore the audit team confirm its reasonability.

Furthermore, when the static investment decreases more than 32.76%, the IRR of the proposed project begins to exceed the benchmark. This cannot be realistic to occur.</td></tr>
<tr><td>operation period (project lifetime in the IRR calculation)</td><td>20</td><td>years</td><td>Feasibility Study report</td><td>Lifetime of the PV modules is 20 years as per the equipment specifications /19/.</td><td>It is reasonable that the project lifetime applies 20 years which is consistent with the lifetime of the main equipments, which complies with the Tool27: Both project internal rate of return (IRR) and equity IRR calculations should reflect the period of expected operation of the underlying project activity (technical lifetime).</td></tr>
<tr><td>Annual net electricity supply</td><td>62,015.75</td><td>MWh</td><td>Feasibility Study report</td><td>As per the Feasibility study report, the annual designed electricity generation of the project is based on long term</td><td>Limited by the solar radiation and generation capacity of solar PV modules, it is unlikely that annual net</td></tr>
</table>

(average)				historical solar radiation data sourced from databases of Meteonorm, SolarGIS, and NASA, which is relatively stable during a long period and can represent the average level during the lifetime of the project. There is not comparable data of similar project in Myanmar.	electricity supplied to the grid increase by 38.28% to enable the IRR of project exceed the benchmark. Therefore, the audit team confirm its reasonability.
Grid tariff (Business tax included)	0.347345	CNY/kWh	Feasibility Study report	Consistent with the power purchase agreement /20/ and "fixed for the entire operation period".	This parameter is consistent with the actual value. Therefore the audit team confirms its reasonability.
Business tax (commercial tax)	5	%	Feasibility Study report	Commercial tax complies with the Commercial Tax Law and Commercial Tax Regulation /27//28/, which is confirmed by public information /30//31/.	This parameter is consistent with the actual value. Therefore the audit team confirm its reasonability.
Corporate Income tax rate	22	%	Feasibility Study report	Consistent with the Myanmar Foreign Investment Law (MFIL) (2012) /29/, which is confirmed by public information /30//32/. It is appropriate and reasonable in the IRR calculation that the year of the exemption of the corporate income tax is considered, which is consistent in the Feasibility Study Report.	This parameter is consistent with the actual value. Therefore the audit team confirm its reasonability.
Scrap value of fixed assets (fair value of the project	5	%	Feasibility Study report	The value applied is common practice which is insignificant in IRR calculation.	This parameter is consistent with the Feasibility Study report and is common practice. Therefore the audit team confirm its reasonability.

activity assets at the end of the assessment period)					
OM costs	602.86	10,000CNY	Feasibility Study report	The Project IRR would reach the benchmark when the annual operation cost decreases by 121.6% which means the operation cost will be negative value to make the IRR reaching benchmark. It is obviously impossible.	The audit team confirm the appropriateness and responsibility of this parameter.
Equity	4,556.35 (25% of the total dynamic investment)	10,000CNY	Feasibility Study report	Consistent with the Financing Proposal signed with the bank /19/	The audit team confirm the appropriateness and responsibility of this parameter.
Long term loan rate	4.3	%	Feasibility Study report	Consistent with the Financing Proposal signed with the bank /19/	The audit team confirm the appropriateness and responsibility of this parameter.

The IRR calculation process in the IRR calculation spreadsheet has been checked and confirmed in accordance of the CDM methodology Tool27. For calculation of financial indicator, all relevant costs and revenues were found to be included in the IRR sheet provided by the PP. All assumptions and estimates used for input values were checked against the relevant sources. The equity IRR value for this project was calculated as 5.47%, which was found to be well below applicable benchmark. The Project Activity cannot be considered as financially feasible as per Tool 01: Tool for demonstration and assessment of additionality.

Sub-step 2d. Sensitivity analysis

A variation of ±10% in the following critical assumptions was considered and presented by the PP in inline to the Tool27. The input parameters that constituted more than 20% or the total revenues/costs have been identified and taken into account in the sensitivity analysis:

IRR Variation	Variation of the parameter indicator needed to reach benchmark
Annual net electricity supplied to the grid	38.28%
Grid tariff	38.28%
Static total investment	-32.76%
Annual O&M Cost	-121.6%

The likelihood of a project activity surpassing the benchmark IRR, in order to ensure the adequacy of the assumptions used in the investment analysis was performed in line with Tool27 in the PD.

Step 3: Barrier analysis

The investment analysis has fully demonstrated and explained the additionality of the project, so step 3 is skipped, which is appropriate.

Step 4: Common Practice Analysis:

Common practice analysis for the project was conducted using methodological Tool24: Common practice version 3.1 and its stepwise approach.

Step 1: The applicable capacity calculated as +/-50% of total design capacity of proposed project activity was 20.14MWp~60.42MWp, which was found to be in line with Tool24 Common practice, version3.1.

Step 2: All similar projects which fulfil the criteria given in Tool24 were identified by the PP considering Myanmar as the applicable geographical area before 04/2022 (signature date of PV module purchase, CDM project start date defined under CDM rules). No similar projects were found in the applicable geographical area meeting the criteria above.

Step 3: It is to be noted that Myanmar is a non-Annex I country and hence the number of ongoing similar project seeking incentives from carbon revenue was checked from websites of other GHG Programs (VCS, GS, CCER, GCC).

It was confirmed that the number of projects neither registered nor submitted for registration (N_{all}) value of 0 power plants is appropriately determined. Therefore, the N_{all}=0.

Step 4: Projects with technologies different to technology applied in the proposed project activity were identified as N_{diff}=0

Step 5: The factor F was found to be calculated in line with Tool 24.

N_{all}=0, N_{diff}=0,

Therefore,

$F=1-N_{diff}/N_{all}=0<0.2$;

$N_{all}- N_{diff}=0<3$

Therefore, the proposed project is not a common practice within the applicable geographical area. Hence, the proposed project is additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

The audit team confirms that the steps taken and equations applied to calculate project emissions, baseline emissions, leakage and emission reductions comply with the requirements of the methodology ACM0002 version 21.0, which are described below.

Baseline emissions

$$BE_y=EG_{PJ,y}\times EF_{grid,CM,y}$$

where

BE_y	Baseline emissions in year y (t CO_2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of “Tool to calculate the emission factor for an electricity system” (tCO_2/MWh)

Calculation of $EG_{PJ,y}$

As the project activity is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, the following applies:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

The baseline emission factor ($EF_{grid,CM,y}$), is calculated as the combination of operating margin ($EF_{grid,OM,y}$) and build margin ($EF_{grid,BM,y}$) factors according to the "Tool to calculate the emission factor for an electricity system".

It is verified that the latest available version for Tool01 "Tool to calculate the emission factor for an electricity system" is version 07.0 and the PP has correctly referred to the same in the section D.3.6 of the PD to determine the baseline grid emission factor.

Step 1. Identify the relevant electricity systems

The project is connected to the Myanmar Power Grid, and the dispatch area, Myanmar Power Grid is delineated as the project electricity system.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional)

Only grid power plants are included in the calculation.

Step 3. Select a method to determine the Operating Margin (OM)

Method (d) Average OM is used.

For the average OM method, the emission factor can be calculated using either of the two following data vintages:

- Ex ante option: A 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD for validation, without requirement to monitor and recalculate the emissions factor during the crediting period, or
- Ex post option: The year in which the project activity displaces grid electricity, requiring the emission factor to be updated annually during monitoring.

The project chose to use the ex-ante vintages and fix the emission factor for the duration of the crediting period. Considering the data availability, data of 3-year between 2014-2016 is used for ex ante calculation.

Step 4. Calculate the Operating Margin emission factor according to the selected method

Because of data availability, as per the *Tool to calculate the emission factor for an electricity system*, Option B of the method Simple OM is employed to calculate the average OM emission factor:

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.
Option B is applicable since:

(i) The necessary data for Option A is not available; and

(ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(iii) Off-grid power plants are not included in the calculation.

Under this option, the average OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system.

Step 5. Calculate the Build Margin (BM) emission factor

The project is located in Myanmar which is an LDC, and the data requirements for the application of Step 5 of the above applied tool cannot be met. Therefore the simplified CM method is applied and combined margin emission factor equals to operation margin emission factor.

Step 6. Calculate the Combined Margin emission factor

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = \omega_{OM} \cdot EF_{grid,OM,y} + \omega_{BM} \cdot EF_{grid,BM,y}$$

Where:

$EF_{grid,BM,,y}$ = Build Margin CO_2 emission factor in year *y* (tCO_2e/MWh)

$EF_{grid,OM,,y}$ = Operating margin CO_2 emission factor in year *y* (tCO_2e/MWh)

ω_{OM} = Weighting of operating margin emissions factor (%)

ω_{BM} = Weighting of build margin emissions factor (%)

Data sources and calculation results of the grid emission factors are confirmed.

For the simplified CM, w_{BM}= 0, w_{OM}= 1, and $EF_{grid,CM,,y} = EF_{grid,OM,,y} =$ 0.35133tCO_2e/MWh.

Project emissions

As per the applied methodology, PE_y = 0 since:

- The project is a solar PV project without fossil fuel consumption
- The project does not involve the integration of a BESS using electricity from the grid or from fossil fuel electricity generators

Leakage

According to the applied methodology, no leakage is considered for the project activity.

Ex-ante emission reduction calculation

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y	Emission reductions in year y (t CO_2e/yr)
BE_y	Baseline emissions in year y (t CO_2/yr)
PE_y	Project emissions in year y (t CO_2e/yr)

The summary of ex ante estimates of emission reductions is shown as follows:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 2023 (18/01/2023 to 31/12/2023, 348 days)	21,617.41	0	0	21,617
Year 2024 (01/01/2024 to 31/12/2024)	22,588.69	0	0	22,588
Year 2025 (01/01/2025 to 31/12/2025)	22,498.79	0	0	22,498
Year 2026 (01/01/2026 to 31/12/2026)	22,408.18	0	0	22,408

Year 2027 (01/01/2027 to 31/12/2027)	22,317.21	0	0	22,317
Year 2028 (01/01/2028 to 31/12/2028)	22,225.21	0	0	22,225
Year 2029 (01/01/2029 to 31/12/2029)	22,132.49	0	0	22,132
Year 2030 (01/01/2030 to 17/01/2030, 17 days)	1,030.63	0	0	1,030
Total	156,818.60	0	0	156,814

Conclusion

The audit team confirms that all data sources and assumptions are appropriate, methodology and referenced tools have been applied correctly, and that calculations are correct, applicable to the project activity, leading to a conservative estimate of the emission reductions. In general, the audit team can confirm the following information:

- All assumptions and data used by the project participants are listed in the PD (version 02 dated 26/02/2024) and/or supporting documents, including their references and sources;

- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PD (version 02 dated 26/02/2024);

- All values used in PD (version 02 dated 26/02/2024) are considered reasonable in the context of the project;

- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;

- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in PD (version 02 dated 26/02/2024).

3.3.7 Methodology Deviations

There is not any methodology deviation.

3.3.8 Monitoring Plan

The project applies the approved monitoring methodology ACM0002 version 21.0 Grid-connected electricity generation from renewable sources. The chosen monitoring methodology is suitable for this project, and the monitoring plan aligns with the monitoring methodologies. The monitoring plan provides a chance for actual measurements of emission reduction achievements. It includes principles and concepts that serve as its foundation, along with operational and monitoring obligations that the project owner must fulfil. These obligations encompass resources involved in the monitoring process, training, support activities, calibration and data collection, quality assurance procedures, data management, and electronic support tools.

The data and parameters available at validation are included in the PD and are provided in compliance with the methodology:

Data / Parameter	Description	Source of data	Value applied	Justification
$EF_{grid,CM,y}$	Baseline combined margin emission factor of Myanmar Power Grid	ACM0002 version 21.0	0.35133tCO_2/MWh	Consistent with methodology

The audit team confirms that the monitoring plan complies with the requirements of the applied methodology and tools. The steps taken to assess whether the monitoring arrangements described in the monitoring plan are feasible within the project design are described below.

Parameter	Description	Monitoring methods
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	Calculated based on the data of $EG_{export,y}$, $EG_{import,y}$, which are measured by electricity meter: $EG_{facility,y} = EG_{export,y} - EG_{import,y}$ $EG_{export,y}$: Electricity supplied to the grid by the project $EG_{import,y}$: Electricity imported from the grid by the project Validation team checked the revised PD and confirmed that the serial number (S/N) and the calibration frequency have been included in the revised PD. Regarding the calibration frequency, Validation team checked the manufacturer's specifications of the electricity meter and confirmed that the specifications requires that the electricity meter test should be performed at periodic intervals according to the valid national regulations. However VVB confirmed that there is no national/industry standards on calibration frequency of electricity meters. Since the electricity meter was manufactured in China by LANDIS, which is a Chinese company, the VVB confirmed that it is appropriate that the calibration frequency follows the calibration standard of China, JJG596-2012 Electrical Meters for Measure Alternating-current Electrical Energy, which requires that the calibration period of the electricity meter with accuracy of 0.2S should not

		be longer than six years. The data will be measured continuously, recorded monthly. All data collected as part of monitoring should be archived electronically and be kept at least for two years after the end of the last crediting period.

By remote interview with the PP, the audit team confirms that the monitoring arrangements described in the monitoring plan are feasible within the project design, and the means of implementation of the monitoring plan are sufficient to ensure the emission reductions achieved by the Project can be reported ex post and verified. The audit team is of the opinion that the monitoring plan complies with the requirements of the methodology.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION OPINION

China Classification Society Certification Co. Ltd. (hereafter referred to as "CCSC") has been commissioned by Myanmar Kindar Solar Power Company Limited to perform the validation of the project activity "Myanmar Kindar Solar Power Plant Project" (hereafter referred to as "the project") on the basis of requirements of Verified Carbon Standard (VCS) Version 4.5, VCS Validation and Verification Manual, and the approved Clean Development Mechanism (CDM) methodology ACM0002 version 21.0, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The validation is an independent 3rd party assessment of the project's baseline, estimated GHG emission reductions or net anthropogenic GHG removals, the monitoring plan and the crediting period using the applicable VCS requirements. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated Verified Carbon Units (VCUs). Verification is the periodic independent review and ex-post determination by a VVB of the monitored reductions in GHG emissions during defined verification period. In carrying out its verification work, the VVB shall ensure that the project complies the applicable CDM and VCS requirements.

The project is eligible under non-AFOLU project. The validation criteria followed the guidance documents provided by VCS and the selected CDM methodology and tool includes:

- VCS Standard, Version 4.5
- VCS Program Guide, Version 4.4
- Methodology ACM0002 version 21.0: Grid-connected electricity generation from renewable sources and related CDM tools.

The project is located in the south of Mandalay Division and 56km northeast to the Meiktila town, Mandalay Division, Myanmar. The purpose of the project is to generate power from clean renewable solar power and supply electricity to the Myanmar Power Grid while contributing to the sustainability of power generation of the Myanmar Power Grid. The project installed 74,592 pieces of solar PV modules, with unit capacity 540 Wp, providing a total capacity of 40.28 MWp. The annual emission reductions in the crediting period is expected 22,402 tCO_2e.

Validation period: From 18/01/2023 to 17/01/2030

Validated GHG emission reductions and removals in the above period:

Year	Estimated GHG emission reductions or removals (tCO_2e)
Year 2023 (18/01/2023 to 31/12/2023, 348 days)	21,617
Year 2024 (01/01/2024 to 31/12/2024)	22,588
Year 2025 (01/01/2025 to 31/12/2025)	22,498
Year 2026 (01/01/2026 to 31/12/2026)	22,408
Year 2027 (01/01/2027 to 31/12/2027)	22,317
Year 2028 (01/01/2028 to 31/12/2028)	22,225
Year 2029 (01/01/2029 to 31/12/2029)	22,132
Year 2030 (01/01/2030 to 17/01/2030, 17 days)	1,030
Total estimated ERs	156,814
Total number of crediting years	7
Average annual ERs	22,402

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full texts
AFOLU	Agriculture, Forestry and Other Land Use
BM	Build Margin
CAR	Corrective Action Request
CCER	China Certified Emission Rrductions
CCSC	China Classification Society Certification Co. Ltd.
CDM	Clean Development Mechanism
CL	Clarification Request
CM	Combined Margin
CO_2	Carbon dioxide
CO_2e	Carbon dioxide equivalent
DOE	Designated Operation Entity
DNA	Designated National Authority
EB	Executive Board
EF	Emission Factor
EIA	Environmental Impact Assessment
ERPA	Emission Reduction Purchase Agreement
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG(s)	Greenhouse gas(es)
GS	Golden Standard
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal Return Rate
MP	Monitoring Plan

PD	Project Description
PDD	Project Design Document
PP	Project Participant
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Voluntary Carbon Unit
VVB	Validation / Verification Body

APPENDIX 2: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

CCS 认证公司
CERTIFICATION

Appendix 9

CERTIFICATE OF COMPETENCE

Date of issue: 04/01/2023

Mr. Li Xingtong

Has been qualified in accordance with *CDM Personnel Competence Requirements and Professional Competence Evaluation Instructions* (CDMI0301) as

■ CDM/VCS validator for Technical Area(s): TA1.1/TA1.2/TA3.1/TA9.2/TA13.1, SS14 for VCS

■ CDM/VCS verifier for Technical Area(s): TA1.1/TA1.2/TA3.1/TA9.2/TA13.1, SS14 for VCS

☐ Technical expert for Technical Area(s):_______________

Tian Wei
CCSC General Manager

CCS 认证公司

Appendix 9

CERTIFICATE OF COMPETENCE

Date of issue: 04/01/2023

Ms. Li Na

Has been qualified in accordance with *CDM Personnel Competence Requirements and Professional Competence Evaluation Instructions* (CDMI0301) as

☐ CDM validator for Technical Area(s):________________

☐ CDM verifier for Technical Area(s):________________

■ Technical expert for Technical Area(s): TA1.2/TA9.2

Tian Wei
CCSC General Manager

CCS 认证公司

Appendix 9

CERTIFICATE OF COMPETENCE

Date of issue: 04/01/2023

Mr. Tang Zhiang

Has been qualified in accordance with *CDM Personnel Competence Requirements and Professional Competence Evaluation Instructions* (CDMI0301) as

■ CDM validator for Technical Area(s): TA1.1/TA1.2/TA.2.1/TA3.1/TA.4.1/TA9.2/TA13.1

■ CDM verifier for Technical Area(s): TA1.1/TA1.2/TA.2.1/TA3.1/TA.4.1/TA9.2/TA13.1

☐ Technical expert for Technical Area(s):________________

Tian Wei
CCSC General Manager

APPENDIX 3: DOCUMENTS REVISED OR REFERENCED

No.	Author	Title	References to the document	Provider
/1/	Myanmar Kindar Solar Power Company Limited	VCS-PD sent to CCSC for validation	Version 01 15/07/2023	PP
/2/	Myanmar Kindar Solar Power Company Limited	Final VCS-PD to request registration	Version 02 26/02/2024	PP
/3/	Myanmar Kindar Solar Power Company Limited	Emission Reductions Calculation Spreadsheet	Version 02 26/02/2024	PP
/4/	Myanmar Kindar Solar Power Company Limited	IRR Calculation Spreadsheet	Version 02 26/02/2024	PP
/5/	CDM-EB	CDM Methodology ACM0002 version 21.0	https://cdm.unfccc.int/methodologies/PAmethodologies/approved	Others
/6/	CDM-EB	Tool01 “Tool for the demonstration and assessment of additionality” (Version 07.0.0)	https://cdm.unfccc.int/Reference/tools/index.html	Others
/7/	CDM-EB	Tool07 “Tool to calculate the emission factor for an electricity system” (Version 07.0)	https://cdm.unfccc.int/Reference/tools/index.html	Others
/8/	CDM-EB	CDM TOOL24 “Common practice” (version 3.1)	https://cdm.unfccc.int/Reference/tools/index.html	Others
/9/	CDM-EB	CDM TOOL27 “Investment analysis” (version 13.0)	https://cdm.unfccc.int/Reference/tools/index.html	Others
/10/	Verra	VCS standard	Version 4.5	Others
/11/	Directorate of Investment and Company Administration, Myanmar	Business license of the project owner	/	PP
/12/	Ministry of Electricity	Letter of Acceptance	15/03/2022	PP

	and Energy, Myanmar			
/13/	Design institute, Kunming Engineering Cooperation Limited	Feasibility Study Report (FSR)	Dated 11/2021	PP
/14/	Design institute, E Guard Environmental Services Co., Ltd.	Environmental Management Plan Report	Dated 05/2021	PP
/15/	PP and contractor	Construction contract	05/2022	PP
/16/	PP and manufacturer JA Solar Technology Co., Ltd.	Main equipment purchase contract – PV module	04/2022	PP
/17/	PP and manufacturer	Main equipment purchase contract – PV racking	04/2022	PP
/18/	PP and manufacturer	Main equipment purchase contract – Inverter	04/2022	PP
/19/	JA Solar Technology Co., Ltd.	Technical specifications of main equipment	/	PP
/20/	PP and grid company	Power purchase agreement (PPA)	/	PP
/21/	PP	Electricity metering records	/	PP
/22/	PP and the bank	Financing Proposal signed with the bank	03/2022	PP
/23/	PP	Training records	/	PP
/24/	PP	Employee lists	/	PP
/25/	PP	Declaration on non-double counting for GHG program	16/08/2023	PP
/26/	Verra	VCS Project Description template	Version 4.2 Dated 21/12/2022	Others
/27/	The State Law and Order Restoration Council, Myanmar	Commercial Tax Law of Myanmar https://www.ird.gov.mm/en/page/617	No. 8/90	Others
/28/	The Ministry of Finance and Revenue, Myanmar	Commercial Tax Regulation of Myanmar https://www.ird.gov.mm/en/page/617	No. 104 / 2012	Others
/29/	The State Law and Order Restoration Council, Myanmar	Myanmar Foreign Investment Law (MFIL) (2012)	No. 21/2012	Others

		https://dica.gov.mm/sites/dica.gov.mm/files/document-files/foreigninvestmentlaw.pdf		
/30/	KPMG	Doing Business in Myanmar	https://assets.kpmg.com/content/dam/kpmg/mm/pdf/2019/05/doing-business-in-myanmar-2019-eng.pdf	Others
/31/	PWC	Myanmar Taxes: Corporate - Other taxes	https://taxsummaries.pwc.com/myanmar/corporate/other-taxes	Others
/32/	PWC	Myanmar Taxes: Corporate - Taxes on corporate income	https://taxsummaries.pwc.com/myanmar/corporate/taxes-on-corporate-income	Others
/33/	Ministry of Electricity and Energy of the Union of Myanmar	Myanmar Energy Statistics 2019	2019	Others

APPENDIX 4: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUEATS AND FORWARD ACTION REQUESTS

Draft report clarifications and corrective action requests by audit team	Summary of project participant response	Audit team conclusion
CL-1 Applicability of the CDM tool-24 shall be justified.	The PD has been revised and applicability of the CDM tool-24 has been justified.	Revised PD has been checked and it is confirmed that CDM tool-24 is applicable. This CL is closed out.
CAR-1 The latest version 13.0 of the CDM tool-27 should be applied.	The latest version 13.0 of the CDM tool-27 is applied in the revised PD. Post tax equity IRR benchmark has been revised to be 20.98% accordingly.	Revision to the PD has been confirmed. This CAR is closed out.
CAR-2 The geographic coordinates of the project are incorrect.	Geographic coordinates of the project have been corrected to be 96°19'19''-96° 19'53''E and 21°10'26''-21°11'03''N.	The location of the project site was verified through remote inspection and the coordinates were verified remote by GPS devices and by checking against Google Maps and Google Earth. This CAR is closed out.